ECN 594: Midterm Exam - SOLUTIONS

1. Short Answer Questions (30 points)

1. (a) (4 points) Profit-maximizing price:

   $\frac{P-MC}{P} = \frac{1}{|\varepsilon|} = \frac{1}{2} \implies P - 30 = 0.5P \implies P^* = \$60$

   (b) (4 points) Interpretation of α:

   α is the price coefficient in dollars: if price goes up by \$1, utility decreases by α dollars.

   (c) (4 points) Adding 5 to all goods (including outside option):

   True. $s_j = \frac{e^{\delta_j+5}}{e^5+\sum_k e^{\delta_k+5}} = \frac{e^5 \cdot e^{\delta_j}}{e^5(1+\sum_k e^{\delta_k})} = \frac{e^{\delta_j}}{1+\sum_k e^{\delta_k}}$

   (d) (4 points) BLP instruments relevance:

   More/closer competitors $\rightarrow$ more competition $\rightarrow$ lower prices.

   (e) (4 points) Higher-priced products typically more elastic:

   True. $\eta_{jj} \approx \alpha p_j$ when s_j small. Since $\alpha < 0$, higher $p_j \rightarrow$ more elastic.

   (f) (3 points) DWL under average cost pricing:

   False. $P = AC > MC \Rightarrow$ consumers with WTP $\in (MC, AC)$ don't buy $\Rightarrow$ DWL > 0.

   (g) (4 points) False CS formula (two goods example):

   Incorrect. Correct formula is $CS = \frac{1}{|\alpha|}\ln(1 + e^{\delta_1} + e^{\delta_2}) = \frac{\ln(1+2e^2)}{|\alpha|} \approx \frac{2.76}{|\alpha|}$.

   The proposed formula gives $\frac{2e^2}{|\alpha|} \approx \frac{14.8}{|\alpha|}$—missing the log.

   (h) (3 points) DWL under perfect price discrimination:

   True. Sells to all with WTP $\geq$ MC $\Rightarrow$ efficient quantity $\Rightarrow$ DWL $= 0$.

2. Demand Estimation (30 points)

2. (a) (5 points) Define δ_{jt} and derive Berry inversion:

   $\delta_{jt} = \beta \cdot \text{mpg}_j + \alpha \cdot p_{jt} + \xi_{jt}$ (part of utility common to all consumers).
   Berry inversion: $\delta_{jt} = \ln(s_{jt}) - \ln(s_{0t})$

   (b) (5 points) Regression equation:

   $\ln(s_{jt}) - \ln(s_{0t}) = \beta \cdot \text{mpg}_j + \alpha \cdot p_{jt} + \xi_{jt}$
   Dependent variable: $\ln(s_{jt}) - \ln(s_{0t})$. Regressors: mpg_j, p_{jt}. Error: ξ_{jt}.

   (c) (5 points) OLS bias:

   $\text{Cov}(p_{jt}, \xi_{jt}) > 0$: high-quality cars command higher prices.
   Upward bias (less negative $\hat{\alpha}$): OLS attributes high demand for expensive cars to low price sensitivity rather than unobserved quality.

   (d) (5 points) Hausman instruments:

If ξ_{jt} is correlated across markets (e.g., brand reputation), then other-market prices are correlated with own-market ξ_{jt}, violating the exclusion restriction.

(e) (5 points) Advantage of demographic interactions:

Allows heterogeneous price sensitivity across consumers, partially addressing IIA: substitution patterns depend on which consumers buy each product.

(f) (5 points) Corolla-Odd vs. Corolla-Even:

Model predicts CS increases: $CS = \frac{1}{|\alpha|}\ln(1 + \sum_j \exp(\delta_j))$ rises when products are added.

Should we believe it? No. The "products" are physically identical—consumers don't care about license plates. This is an example of the "red-bus blue-bus" problem.

3. Two-Part Tariffs (20 points)

3. (a) (5 points) Profit as a function of F and p:

$q = 20 - 2p$, choke price $p = 10$.
$CS(p) = \frac{1}{2}(20 - 2p)(10 - p) = (10 - p)^2$, so max fee $F = (10 - p)^2$.
$\Pi = 100\left[(10 - p)^2 + (p - 2)(20 - 2p)\right]$

(b) (5 points) Show $p^* = c = 2$:

$\frac{d\pi}{dp} = -2(10 - p) + (20 - 2p) - 2(p - 2) = 4 - 2p$
$4 - 2p = 0 \Rightarrow p^* = 2 = c$

(c) (10 points) Comparing two pricing strategies:

At $p = 0$: $CS_H = \frac{1}{2}(10)(10) = 50$, $CS_L = \frac{1}{2}(5)(5) = 12.5$.

(a) $F = 50$: Heavy join ($CS_H = 50 \geq F$), light don't ($CS_L = 12.5 < F$). Profit $= 50 \times 50 = \$2,500$.

(b) $F = 12.5$: Both join. Profit $= 150 \times 12.5 = \$1,875$.

Strategy (a) is better.

4. Self-Selection (20 points)

4. (a) (4 points) Perfect price discrimination:

Businesses: $P_P = \$400$. Students: $P_B = \$100$.
Profit $= 40(370) + 60(70) = \$19,000$

(b) (4 points) IC constraint for businesses:

$400 - P_P \geq 150 - P_B \implies P_P - P_B \leq 250$

(c) (6 points) Why $P_P = \$380$, $P_B = \$100$ is not optimal:

Business surplus from Pro: $400 - 380 = 20$. From Basic: $150 - 100 = 50$.
Since $20 < 50$, businesses prefer Basic. IC violated: $P_P - P_B = 280 > 250$.

(d) (6 points) Self-selection prices:

Binding: Student IR ($P_B = 100$) and Business IC ($P_P = P_B + 250 = 350$).
Check student IC: $100 - 100 = 0 \geq 120 - 350 = -230$ Profit $= 40(320) + 60(70) = \$17,000$